The objective of my project was to:

- Build a **Star Schema data model**
- Define **primary and foreign key relationships**
- Apply correct **cardinality and filter direction**
- Handle **inactive relationships**
- Create **hierarchies for better data navigation**
- Validate the model using **Matrix visuals**

Dataset Description

In this project, I worked with multiple datasets:

Fact Tables

- **Sales_Fact** → Main transactional table containing Revenue, Quantity, Discount
- **Returns_Fact** → Contains return-related data linked to sales

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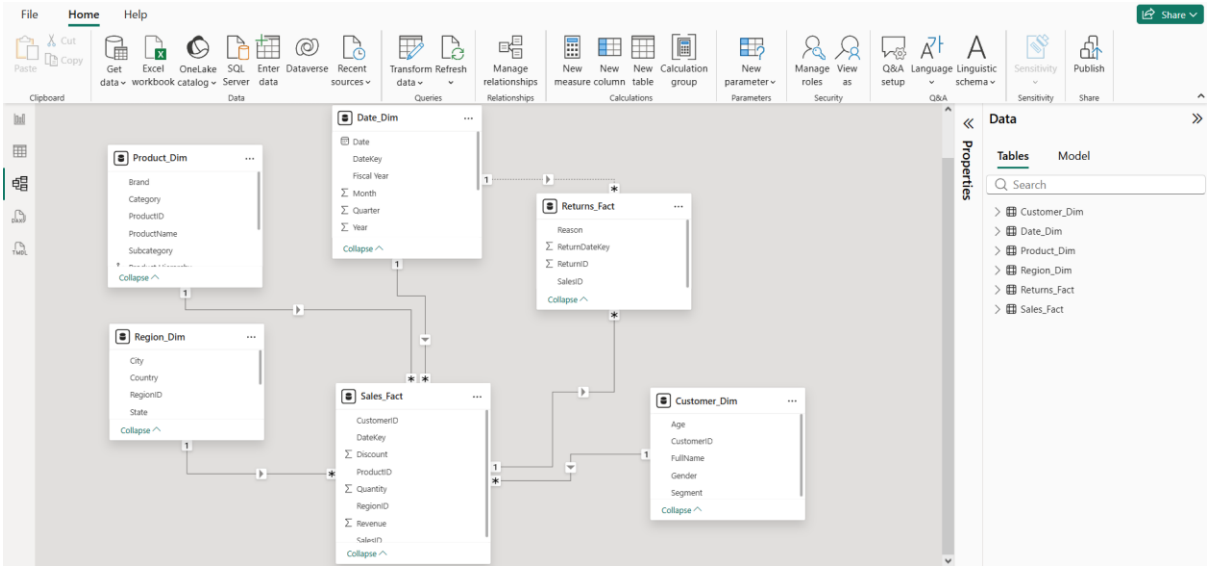
Dimension Tables

- **Customer_Dim** → Customer details
- **Product_Dim** → Product information
- **Region_Dim** → Location details
- **Date_Dim** → Time-based data

I imported all datasets using Power Query and performed the following steps:

- Removed null and duplicate values
- Renamed columns for consistency
- Assigned correct data types:
    - IDs → Whole Number
    - Revenue → Decimal
    - Date → Date format



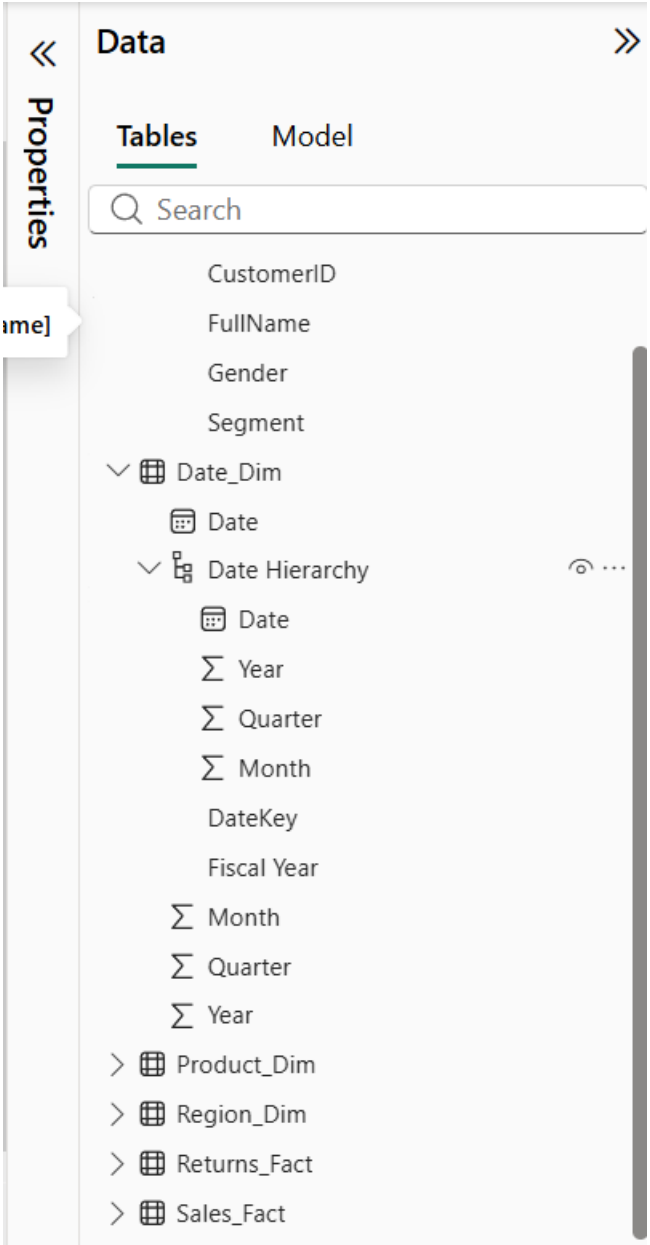
I created the following relationships:

- Sales → Customer (Many to One)
- Sales → Product (Many to One)
- Sales → Region (Many to One)
- Sales → Date (Many to One)
- Returns → Sales (Many to One)
- Returns → Date (**Inactive Relationship**)

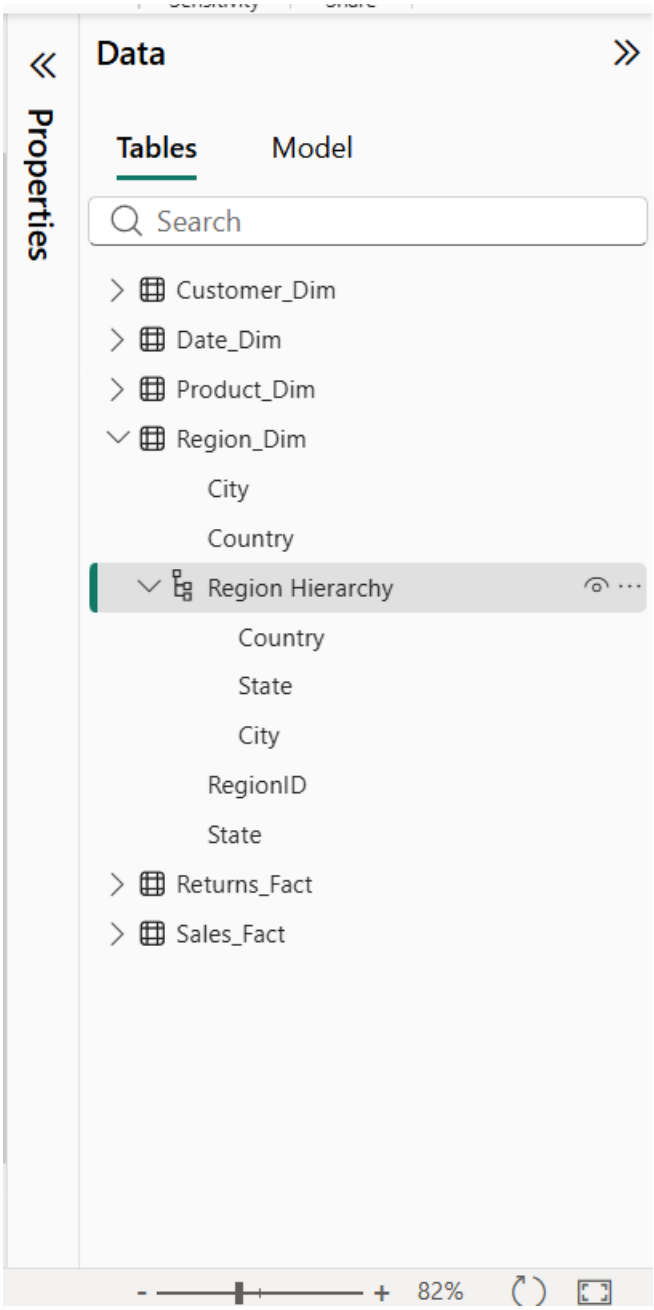
Hierarchies

I created the following hierarchies:

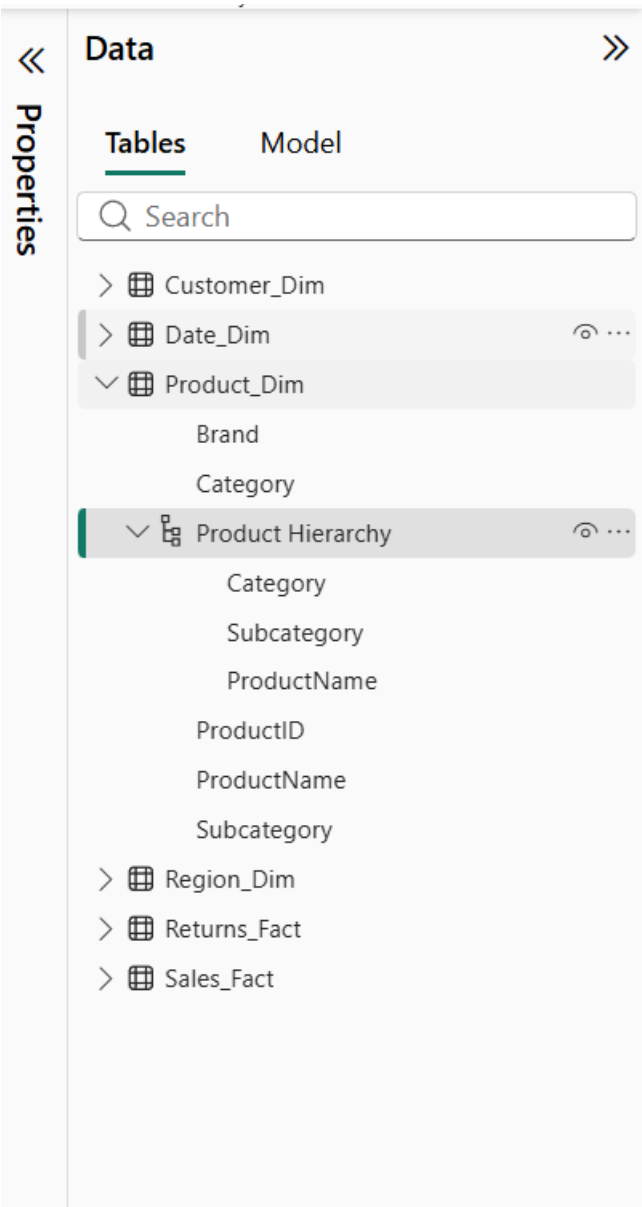
- Date → Year → Quarter → Month → Date

- 

- Region → Country → State → City

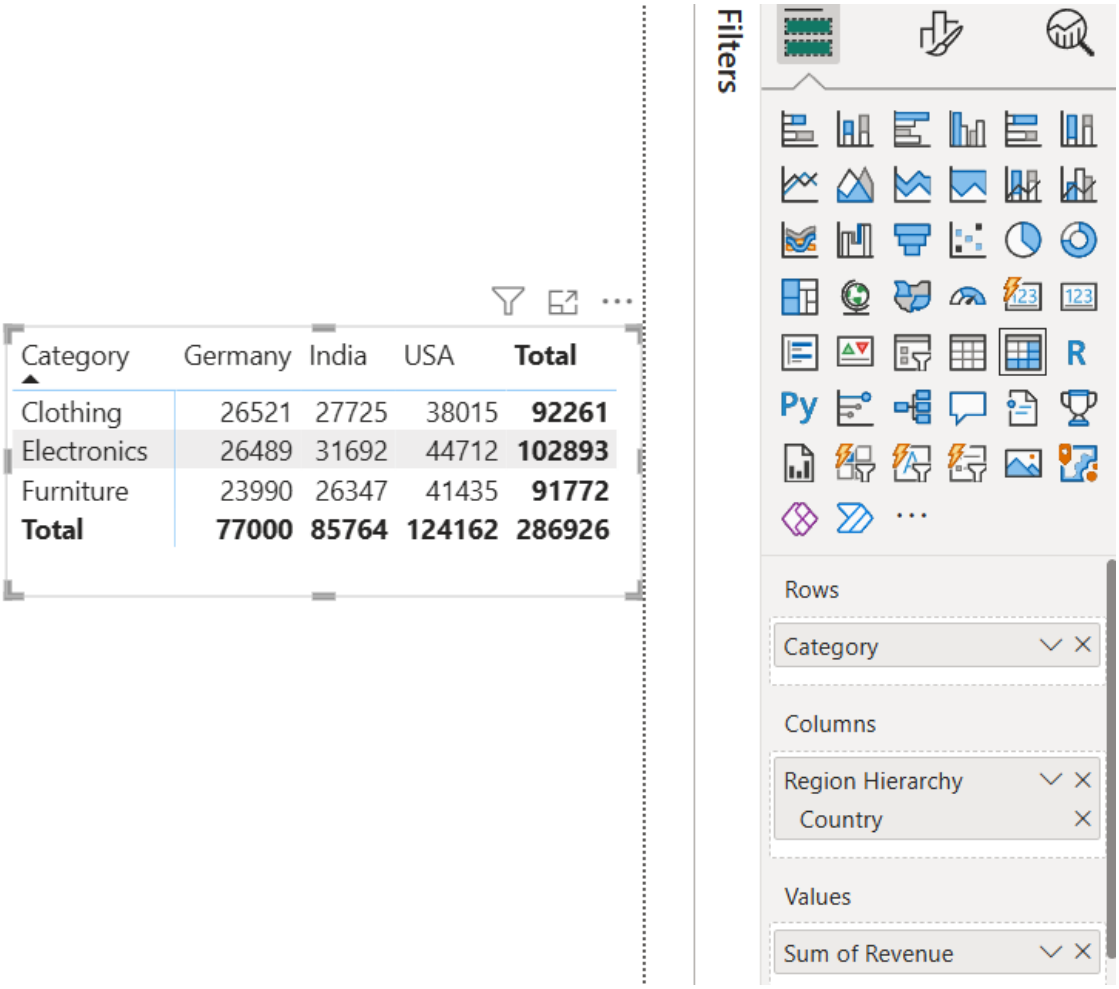
- 

- Product → Category → Subcategory → ProductName

- 

- **Category vs Country**

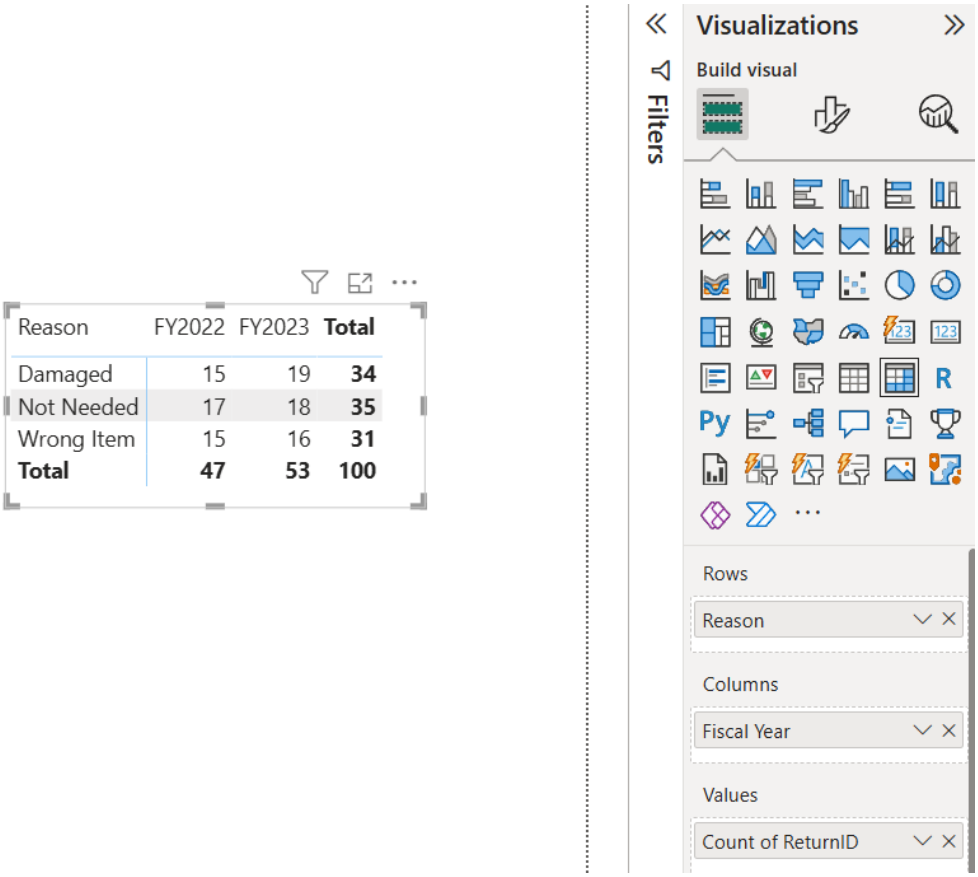
  "Electronics has highest revenue **102,893**, and USA contributes **44,712**, showing correct Product–Region filtering."



Category	Germany	India	USA	Total
Clothing	26521	27725	38015	**92261**
Electronics	26489	31692	44712	**102893**
Furniture	23990	26347	41435	**91772**
Total	**77000**	**85764**	**124162**	**286926**

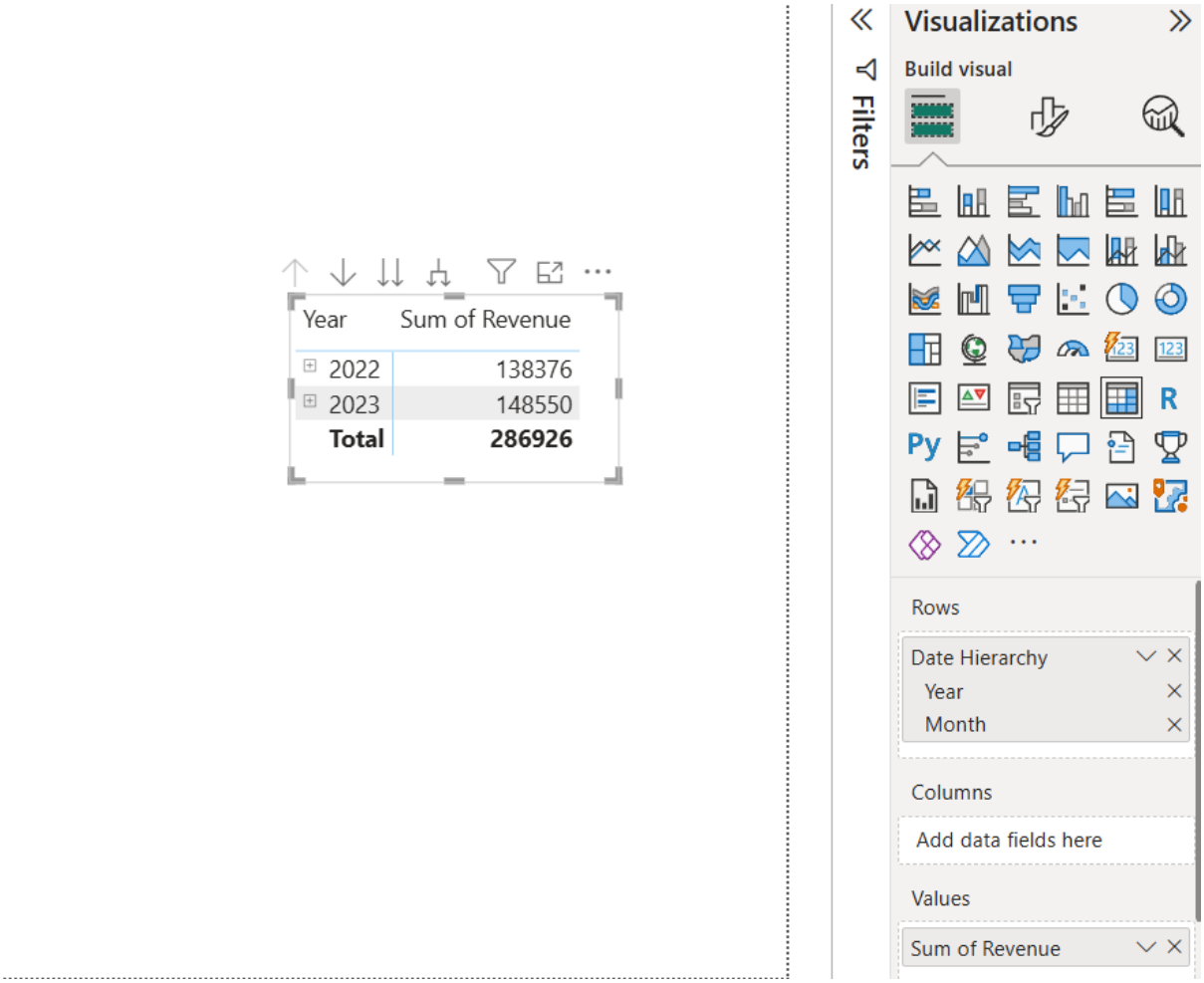
- **Return Reason vs Fiscal Year**

  "Total returns are **100**, with 'Not Needed' highest at **35**, validating Returns–Date relationship."



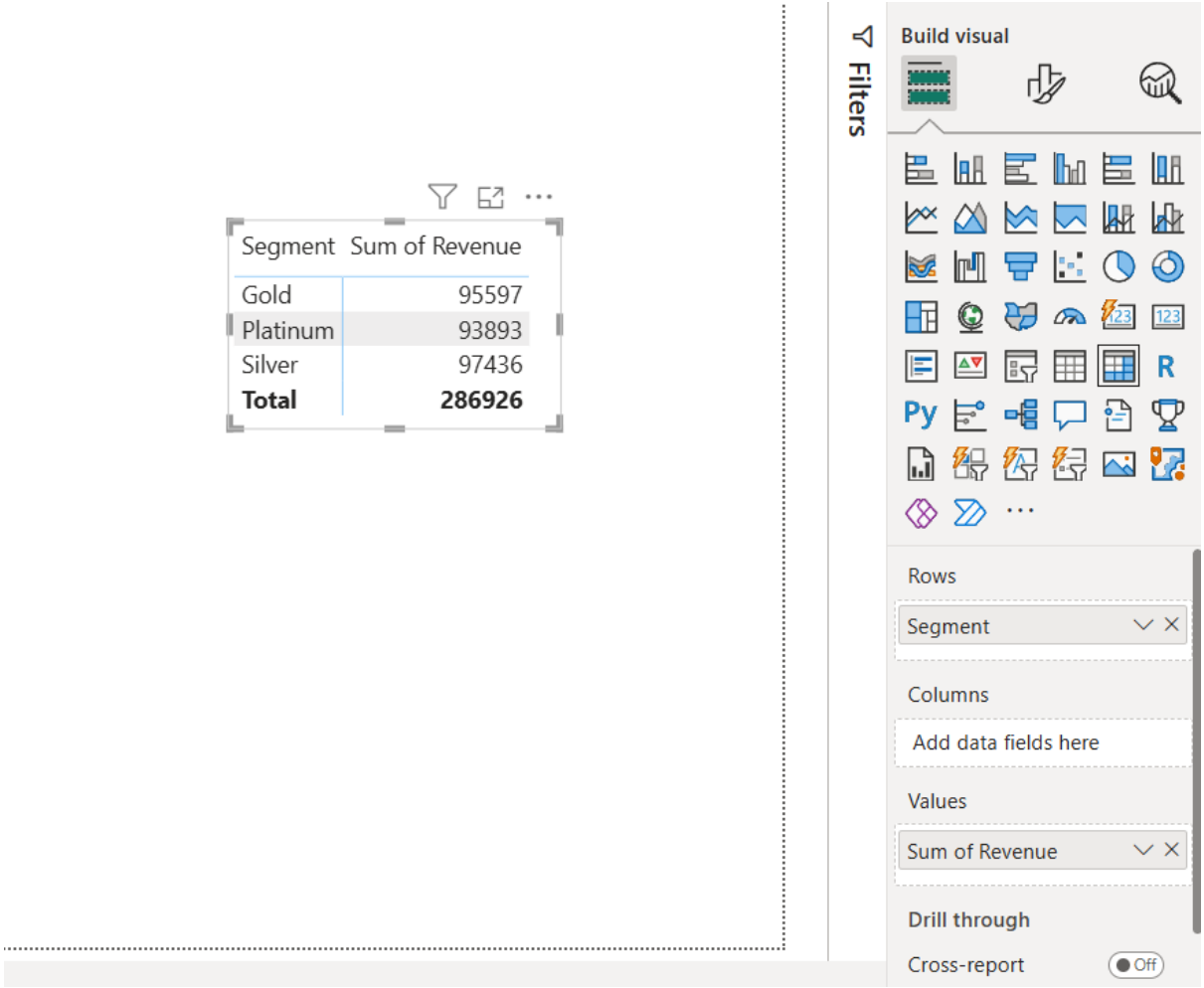
- **Year vs Revenue**

"Revenue increased from **138,376 in 2022** to **148,550 in 2023**, confirming time-based aggregation."



- **Segment vs Revenue**

"Silver segment leads with **97,436**, followed by Gold and Platinum, validating Customer filtering."



- **Country vs Year**

"USA has highest total **124,162**, and overall revenue is **286,926**, showing correct multi-dimensional filtering."

Country	2022	2023	Total
Germany	30794	46206	**77000**
India	43112	42652	**85764**
USA	64470	59692	**124162**
Total	**138376**	**148550**	**286926**

Filters

Visualizations

Build visual

Rows

Region Hierarchy
- Country
- State
- City

Columns

Year

Values

Sum of Revenue